

Playwright with TypeScript

Week 5 – Home Assignment



Program	Playwright with TypeScript
Module	TypeScript Advanced Types
Week	Week 5
Assignment	Practice Type Alias, Union and Intersection in TypeScript (Product and Order)

1. Objective

To practice and understand how to use Type Alias, Union types and Intersection types in TypeScript by solving a real-time style problem.

2. Problem Statement

Write TypeScript programs to manage a product and order system using Type Alias, Union and Intersection types.

3. Requirements

- Create a Type Alias called Product with the following properties:
 - id (number)
 - name (string)
 - price (number)
 - category (string)
- Create a Union type called ProductStatus with values: "In Stock" | "Out of Stock" | "Discontinued" and use it in the Product type.
- Create a new object of the Product type and print the details.
- Update the product to use the ProductStatus union type for a status property and create objects with each of the possible status values.
- Create two separate type aliases:
 - ProductDetails (id, name, price, status)
 - InventoryDetails (quantity, location)
- Create an Intersection type called ProductInfo by combining ProductDetails and InventoryDetails.
- Create an object of ProductInfo and print all the properties.
- Try updating the price and quantity values and print the updated object.
- Add meaningful comments for each step and use proper TypeScript types.

4. Sample Test Data and Expected Output

Sample Data	Expected Output
<pre>// For Type Alias and Union id: 1 name: "Laptop" price: 65000 category: "Electronics" status: "In Stock" "Out of Stock" "Discontinued" // For Intersection id: 2 name: "Mobile Phone" price: 25000 status: "In Stock" quantity: 50 location: "Chennai"</pre>	<pre>// Product object with Union type { id: 1, name: 'Laptop', price: 65000, category: 'Electronics', status: 'In Stock' } // ProductInfo object using Intersection type { id: 2, name: 'Mobile Phone', price: 25000, status: 'In Stock', quantity: 50, location: 'Chennai' }</pre>

5. Technical Guidelines

- Use TypeScript (.ts file).
- Use type alias, union and intersection as per the requirements.
- Use console.log() to display the output.
- Use meaningful variable names and proper formatting.
- Add comments to explain the code.

6. Learning Outcome

After completing this assignment, you should be able to:

- Create and use type alias in TypeScript.
- Implement union types and use them in a real-time scenario.
- Create and use intersection types by combining multiple types.
- Work with structured data using advanced type features in TypeScript.